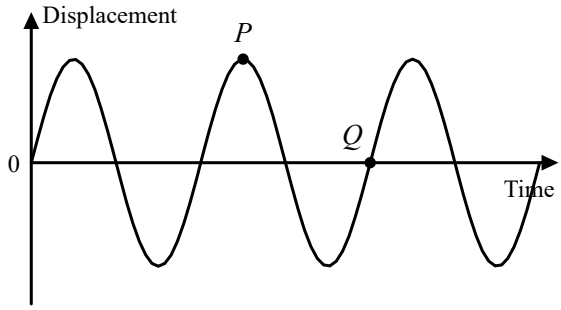
STPM 2013 PHYSICS 960/3 (U3)

Questions and Answers

Values of constants: $h = 6.63 \times 10^{-34}$ J s, $c = 3.00 \times 10^{8}$ m s^{-1}, $e = 1.60 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg, 1 u $= 1.66 \times 10^{-27}$ kg.

Section A [15 marks]

1. The variation of displacement with time for an object undergoing simple harmonic motion is shown in the graph below.



Which statement about the motion of the object is **not** true?

A At *P*, the potential energy is maximum.
B At *P*, the resultant force is maximum.
C At *Q*, the acceleration is maximum.
D At *Q*, the speed is maximum.

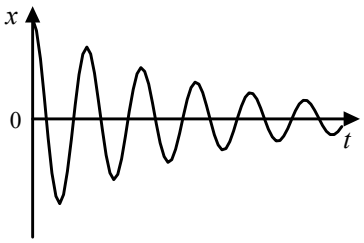
Answer: C

Q is the equilibrium position, where the displacement and hence the acceleration are zero. Statement C is not true.

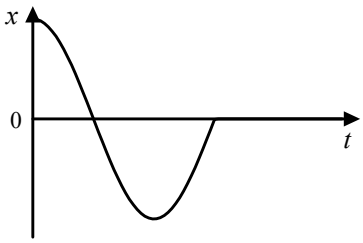
Explanation: At *P* (maximum displacement) the potential energy and restoring force are maximum; at *Q* the speed is maximum.

2. Which displacement *x* against time *t* graph represents motion of an oscillatory system which experiences overdamped oscillation?

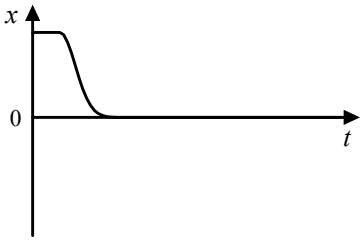
A



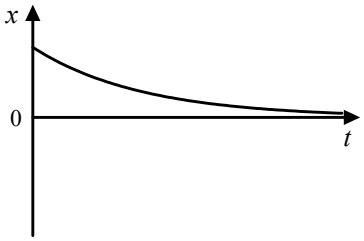
B



C



D

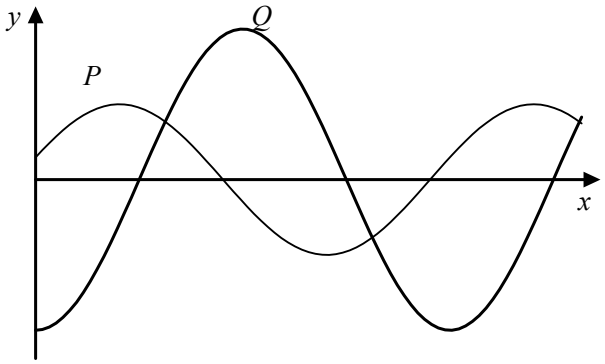


Answer: D

Overdamping: the system returns slowly to equilibrium without oscillating (D).

Explanation: A is underdamped; C (quick return, no oscillation) is critical damping.

3. Two progressive waves P and Q travelling in positive-x direction is shown in the graph below.



Which statement is true of the two progressive waves?

A The wave Q leads P by $\frac{\pi}{2}$ radian.

B The waves P and Q are π radian out-of-phase.

C The frequency of the wave Q is twice than that of P.

D The intensity of the wave P is higher than that of Q.

Answer: A

Both waves have the same wavelength and speed, hence the same frequency; Q has the larger amplitude, so it has the greater intensity.

From the graph the two waves are displaced by a quarter of a wavelength, with Q leading P by $\pi/2$ (A).

Explanation: A shift of $\lambda/4$ corresponds to a phase difference of $\pi/2$.

4. Two loud speakers which are placed a distance apart and facing each other emit sounds of the same frequency. When a small microphone is moved along the line joining them, maximum intensities are detected at intervals of 10 cm. If the speed of sound in air is 350 m s^{-1}, what is the frequency of the sound?

A 875 Hz B 1750 Hz C 3500 Hz D 7000 Hz

Answer: B

Maxima (antinodes) are $\frac{\lambda}{2}$ apart: $\lambda = 0.20$ m $\Longrightarrow f = \frac{350}{0.20} = 1750$ Hz

Explanation: The two speakers set up a standing wave between them.

5. A closed pipe has a length of 4.8 cm. If the fundamental frequency is 1500 Hz, what is the frequency for the first overtone and the end-correction of the pipe?
[The speed of sound in air is 300 m s^{-1}.]

	Frequency	*End-correction*
A	3000 Hz	5.2 cm
B	3000 Hz	0.2 cm
C	4500 Hz	5.2 cm
D	4500 Hz	0.2 cm

Answer: D

Closed pipe: first overtone = $3f_0 = 4500$ Hz

$\lambda_0 = \frac{300}{1500} = 0.20$ m; $L + e = \frac{\lambda}{4} = 5.0$ cm $\Longrightarrow e = 0.2$ cm

Explanation: A pipe closed at one end has only odd harmonics.

6. If the intensity level of a sound at a point is increased by 3 dB, what is the ratio of the new intensity to the initial intensity?

A 0.3 B 0.5 C 2 D 3

Answer: C

$10 \log \frac{I_2}{I_1} = 3 \Longrightarrow \frac{I_2}{I_1} = 10^{0.3} \approx 2$

Explanation: Doubling the intensity adds about 3 dB.

7. When an object is placed 5 cm from a curved mirror, an image which is twice the size of the object is formed behind the mirror. Which is true of the curved mirror?

A It is a convex mirror of radius of curvature 10 cm.
B It is a concave mirror of radius of curvature 10 cm.
C It is a convex mirror of radius of curvature 20 cm.
D It is a concave mirror of radius of curvature 20 cm.

Answer: D

Virtual, magnified image $\Longrightarrow$ concave mirror; $v = -10$ cm: $\frac{1}{f} = \frac{1}{5} - \frac{1}{10} \Longrightarrow f = 10$ cm, $r = 20$ cm

Explanation: A convex mirror always gives a diminished image.

8. Which statement is **not** true of Huygens's principle?

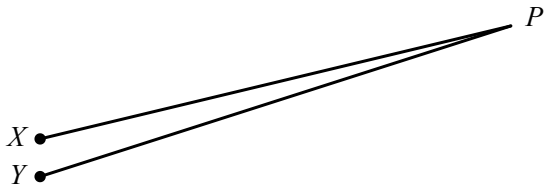
A Every point on a wave front is a source of secondary wavelets.
B The new wave front is the common tangent of all the secondary wavelets.
C The wave front is parallel to the direction of wave propagation at every point.
D The wavelets spread out in the forward direction at the speed of the wave itself.

Answer: C

A wavefront is perpendicular (not parallel) to the direction of propagation. Statement C is not true.

Explanation: Rays are normal to the wavefronts.

9. Two coherent sources X and Y each produce light of wavelength λ and amplitude A is shown in the diagram below.



A constructive interference of m order occurs at P. If the intensity of light source for X and Y at P is I_o, which is true of the path difference ($YP - XP$), the resultant amplitude and the resultant intensity of the light at P?

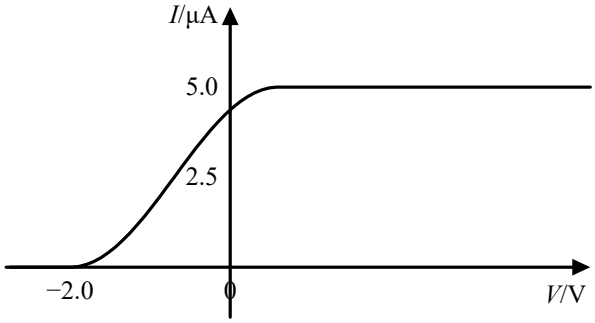
	(YP − XP)	*Resultant amplitude*	*Resultant intensity*
A	$m\lambda$	$2A$	$2I_o$
B	$m\lambda$	$2A$	$4I_o$
C	$(m + 1)\lambda$	A	$2I_o$
D	$(m + 1)\lambda$	A	$4I_o$

Answer: B

Constructive interference: path difference $m\lambda$; resultant amplitude $2A$; intensity $\propto$ amplitude^2, so $4I_o$.

Explanation: The energy is redistributed: dark fringes elsewhere balance the bright ones.

10. The variation of photoelectric current I with potential difference V across a photocell is shown in the graph below.



What is the maximum speed of the photoelectrons emitted?

A 5.9×10^5 m s^{-1} B 8.4×10^5 m s^{-1} C 1.2×10^6 m s^{-1} D 1.7×10^6 m s^{-1}

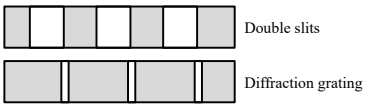
Answer: B

$$V_s = 2.0 \text{ V}: \frac{1}{2}mv^2 = eV_s \Longrightarrow v = \sqrt{\left(\frac{2(1.60 \times 10^{-19})(2.0)}{9.11 \times 10^{-31}}\right)} = 8.4 \times 10^5 \text{ m s}^{-1}$$

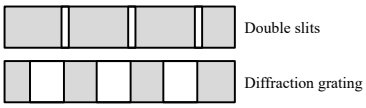
Explanation: Read the stopping potential where the current becomes zero.

11. A double slits and a diffraction grating which have equal slit separations are irradiated with a monochromatic light of same wavelengths. Which diagram shows the correct interference patterns formed?

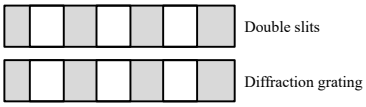
A



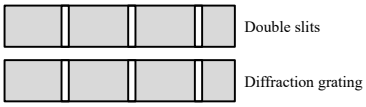
B



C



D



Answer: A

With the same slit separation, the bright fringes are at the same positions; the grating gives much sharper (narrower) and brighter lines with wide dark regions between them (A).

Explanation: Many slits make the maxima narrower and brighter.

12. In a hydrogen atom, the energy E_n of the nth state is given by $E_n = -\frac{k}{n^2}$, where k is a constant. An electronic transition from the first excited state to the ground state emits a photon of wavelength λ. What is the wavelength of the emitted photon if the electronic transition is from the nth state to the ground state?

A $\frac{\lambda}{n^2}$ B $\frac{n^2\lambda}{n^2-1}$ C $\frac{3\lambda}{n^2-1}$ D $\frac{3n^2\lambda}{4(n^2-1)}$

Answer: D

$$\frac{hc}{\lambda} = k\left(1 - \frac{1}{4}\right) = \frac{3k}{4};\ \frac{hc}{\lambda_n} = k\left(1 - \frac{1}{n^2}\right)$$

$$\lambda_n = \frac{3n^2\lambda}{4(n^2 - 1)}$$

Explanation: Take the ratio so that k and hc cancel.

13. Which statement is true of de Broglie hypothesis?

A Particles have wave properties.
B Particles have magnetic properties.
C Particles have electrical properties.
D Particles have wave and electrical properties.

Answer: A

De Broglie proposed that moving particles have wave properties, with $\lambda = h/p$.

Explanation: This was confirmed by electron diffraction.

14. An atom of element ${}^{A}_{Z}X$ has a mass M_x. The masses of proton, neutron and electron are m_p, m_n and m_e respectively, and the speed of light in a vacuum is c. What is the binding energy of the nucleus ${}^{A}_{Z}X$?

A $[Zm_p + (A - Z)m_n - (M_x + m_e)]\,c^2$
B $[Zm_p + (A - Z)m_n - (M_x - m_e)]\,c^2$
C $[Zm_p + (A - Z)m_n - (M_x - Zm_e)]\,c^2$
D $[Zm_p + (A - Z)m_n - (M_x + Zm_e)]\,c^2$

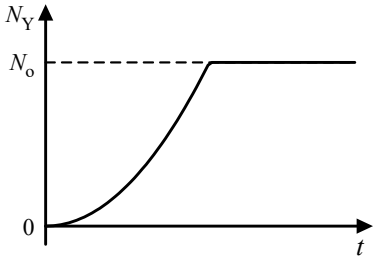
Answer: C

$$\text{Nuclear mass} = M_x - Zm_e;\ E_B = [Zm_p + (A - Z)m_n - (M_x - Zm_e)]c^2$$

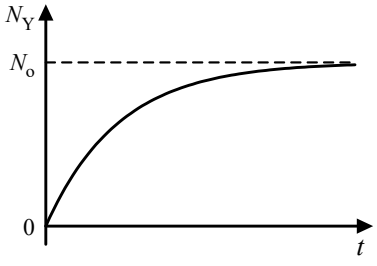
Explanation: Remove the mass of the Z electrons from the atomic mass.

15. A radioactive element X decays to a stable element Y by emitting a radioactive particle. Starting with a sample of nucleus X about N_o, and without nucleus Y, which graph shows the variation of nucleus number N_Y with time t?

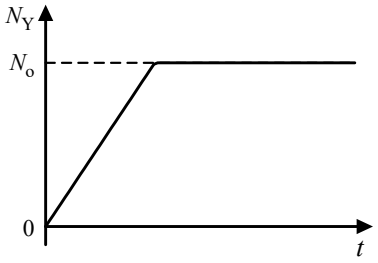
A



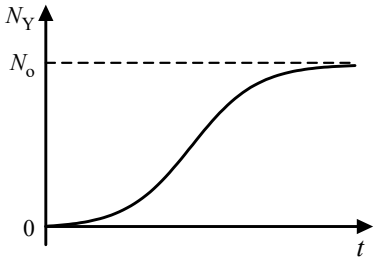
B



C



D



Answer: B

$$N_Y = N_o - N_X = N_o(1 - e^{-\lambda t})$$

It rises quickly at first and approaches N_o exponentially (B).

Explanation: The number of daughter nuclei grows as the parent decays.

Section B [15 marks]

16. (a)

(i) State **two** characteristics of the reflected light when unpolarised light is incident with Brewster's angle onto a glass surface. [2]

Answer

The reflected light is completely plane polarised (vibrations parallel to the surface)	1
The reflected and refracted rays are perpendicular to each other	1
	2

Explanation: At Brewster's angle $\tan \theta_B = n$.

(ii) Polarisation occurs when sunlight falls onto a glass panel at the side of a building. Determine Brewster's angle of the polarised light.
[The refractive index of glass is 1.52.] [2]

Answer

$\tan \theta_B = n = 1.52$	1
$\theta_B = 56.7°$	1
	2

Explanation: Brewster's law.

(b)

(i) Two polarising sheets are placed together with their axes of polarisation making an angle of 35°. Unpolarised light passes through them. Determine the percentage of the light intensity transmitted through both polarisers. [2]

Answer

$I = \frac{I_0}{2} \cos^2 35°$ 1

$= 0.336 I_0$: 33.6% transmitted 1

2

Explanation: The first sheet halves unpolarised light.

(ii) State **two** advantages of wearing a pair of Polaroid sunglasses during an afternoon stroll when the sun is bright hot. [2]

Answer

Reduce glare: light reflected from horizontal surfaces is partly polarised and is blocked 1

Reduce the intensity of light entering the eyes, making vision more comfortable 1

2

Total for question **8**

Explanation: The axis of Polaroid sunglasses is vertical.

17. (a) State **two** characteristics of X-ray. [2]

Answer

• Electromagnetic waves of very short wavelength (about 10^{-10} m), travel at the speed of light 1

• Highly penetrating, ionise gases, not deflected by electric or magnetic fields, affect photographic film 1

2

Explanation: Any two characteristics.

(b) When a monochromatic X-ray beam of wavelength 8.3×10^{-11} m falls on the surface of a sodium chloride crystal, the first order Bragg diffraction occurs at a glancing angle of 8.5°.

(i) Calculate the separation of the atomic plane. [3]

Answer

$2d \sin \theta = n\lambda$ with $n = 1$ 1

$d = \dfrac{8.3 \times 10^{-11}}{2 \sin 8.5°}$ 1

$d = 2.81 \times 10^{-10}$ m 1

3

Explanation: The glancing angle is measured from the crystal planes.

(ii) Determine the maximum order of diffraction that can be obtained. [2]

Answer

$n \leq \dfrac{2d}{\lambda} = \dfrac{2(2.81 \times 10^{-10})}{8.3 \times 10^{-11}} = 6.8$ 1

Maximum order $n = 6$ 1

2

Total for question **7**

Explanation: $\sin \theta$ cannot exceed 1.

Section C [30 marks]

18. (a) State **two** differences between a progressive wave and a standing wave. [2]

Answer

• A progressive wave transfers energy; a standing wave does not 1

• Progressive: all particles have the same amplitude; standing: amplitude varies from zero at nodes to maximum at antinodes 1

2

Explanation: Also: the waveform of a standing wave does not move.

(b)

(i) Explain the principle of superposition for progressive waves. [2]

Answer

When two or more waves meet, the resultant displacement at any point is the vector sum of the individual displacements — 1

After crossing, each wave continues unchanged — 1

2

Explanation: This explains interference and standing waves.

(ii) Explain the occurrence of resonance in a hollow open-ended pipe. [2]

Answer

Sound reflected at the open ends sets up a standing wave in the air column with antinodes at both ends — 1

Resonance (a loud sound) occurs when the driving frequency equals a natural frequency of the pipe: $L = n\lambda/2$ — 1

2

Explanation: The fundamental of an open pipe has $\lambda = 2L$.

(c) Two progressive waves are represented by $y_1 = 0.2 \sin (100\pi t + 4\pi x)$ and $y_2 = 0.2 \sin (100\pi t - 4\pi x)$.

(i) Derive an expression for the resultant wave as a result of superposition. [3]

Answer

$y = y_1 + y_2 = 0.2[\sin (100\pi t + 4\pi x) + \sin (100\pi t - 4\pi x)]$ — 1

$= 2(0.2) \sin (100\pi t) \cos (4\pi x)$ — 1

$y = 0.4 \cos (4\pi x) \sin (100\pi t)$ — 1

3

Explanation: Use $\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$.

(ii) State the changes in the profile of the resultant wave as compared to the original waves. [2]

Answer

The resultant is a standing wave: its profile does not move along the string — 1

The amplitude varies with position, from 0 (nodes) to 0.4 m (antinodes), twice that of each original wave — 1

2

Explanation: The frequency is unchanged.

(iii) Calculate the period of the resultant wave. [2]

Answer

$T = \frac{2\pi}{\omega} = \frac{2\pi}{100\pi}$ 1

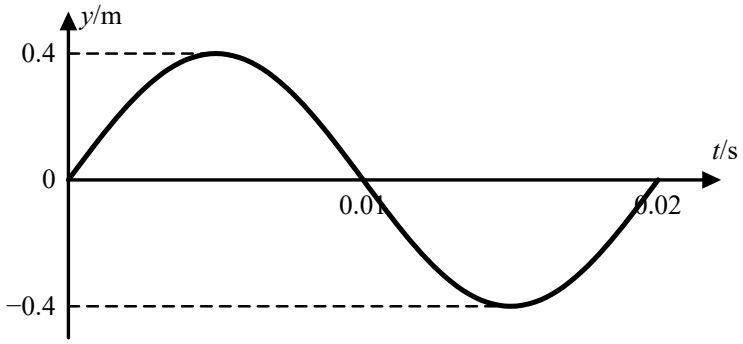
$T = 0.02$ s 1

2

Explanation: The same period as each progressive wave.

(iv) Sketch the displacement against time graph of the resultant wave for one cycle of a particle at $x = 0$. [2]

Answer



Sine curve starting at 0, amplitude 0.4 m, one cycle in 0.02 s 2

2

Total for question **15**

Explanation: At $x = 0$, $\cos 0 = 1$, so the particle is at an antinode.

19. (a) A diver is at 1.75 m from a flat glass window of an aquarium. A spectator looks into the aquarium through the window. Neglecting the thickness of the glass window, determine the distance of the diver from the window as seen by the spectator.
[The refractive index of the water is 1.33.] [3]

Answer

Apparent distance $= \frac{\text{real distance}}{n}$ 1

$= \frac{1.75}{1.33}$ 1

= 1.32 m from the window 1

3

Explanation: Light from the diver bends away from the normal as it leaves the water, so the diver appears closer.

(b) An object is placed in front of a converging lens with a focal length of 2.45 cm. An image is formed at a distance of 13.5 cm from the object. Determine the position of the lens from the object if the image is

(i) real, [3]

Answer

Real image: $u + v = 13.5$ cm — 1

$\frac{1}{u} + \frac{1}{13.5 - u} = \frac{1}{2.45} \Rightarrow u^2 - 13.5u + 33.1 = 0$ — 1

$u = 3.22$ cm or 10.3 cm from the object — 1

3

Explanation: Two positions of the lens give a sharp real image (they are interchangeable).

(ii) virtual. [3]

Answer

Virtual image on the same side: $|v| = u + 13.5$ — 1

$\frac{1}{u} - \frac{1}{u + 13.5} = \frac{1}{2.45} \Rightarrow u^2 + 13.5u - 33.1 = 0$ — 1

$u = 2.12$ cm — 1

3

Explanation: The object is inside the focal length (magnifying glass).

(c) When an object of height 5.00 cm is projected through a thin lens onto a screen, it forms an image of 1.25 m high. The distance between the object and screen is 3.00 m.

(i) Calculate the focal length of the lens. [3]

Answer

$m = \frac{125}{5.00} = 25 \Rightarrow v = 25u$; $u + v = 300$ cm — 1

$u = 11.5$ cm, $v = 288$ cm — 1

$f = \frac{uv}{u + v} = 11.1$ cm — 1

3

Explanation: A real image on the screen needs a converging lens.

(ii) If the lens is made of glass ($n = 1.52$) and has a plano-convex shape, determine the radius of its curvature. [3]

Answer

$\frac{1}{f} = (n-1)\frac{1}{r}$ (plano-convex) 1

$r = (1.52 - 1)(11.1)$ 1

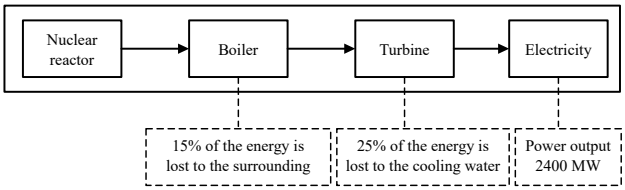
$r = 5.77$ cm 1

3

Total for question **15**

Explanation: The flat surface has an infinite radius.

20. A schematic diagram for a nuclear power station is shown in the diagram below.



The following reaction represents a nuclear fission in the above nuclear reactor.

$^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{141}_{56}Ba + {}^{92}_{36}Kr + 3{}^{1}_{0}n$

[Mass of $^{235}_{92}U = 235.04392$ u, mass of $^{1}_{0}n = 1.00867$ u, mass of $^{141}_{56}Ba = 140.91408$ u, mass of $^{92}_{36}Kr = 91.92500$ u and 1 u ≡ 934 MeV.]

(a) Explain the functions of

(i) a moderator, [2]

Answer

The moderator (e.g. graphite or water) slows down the fast neutrons released in fission 1

Slow (thermal) neutrons are much more likely to cause fission of U-235, sustaining the chain reaction 1

2

Explanation: Neutrons lose energy by colliding with light nuclei.

(ii) the control rods in a nuclear reactor. [2]

Answer

Control rods (e.g. boron or cadmium) absorb neutrons 1

Moving them in or out controls the number of neutrons and so the rate of fission (power output) 1

2

Explanation: Fully inserted rods shut the reactor down.

(b) Determine the energy released for one nuclear fission reaction. [3]

Answer

$\Delta m = (235.04392 + 1.00867) - (140.91408 + 91.92500 + 3 \times 1.00867) = 0.18750$ u — 1

$E = (0.18750)(934) = 175$ MeV — 1

$E = (175 \times 10^{6})(1.60 \times 10^{-19}) = 2.80 \times 10^{-11}$ J — 1

3

Explanation: Use 1 u ≡ 934 MeV as given.

(c) Calculate the total power generated by the reactor. [3]

Answer

Energy lost: 15% + 25% = 40%, so the output 2400 MW is 60% of the total — 1

Total power $= \frac{2400}{0.60}$ — 1

$= 4000$ MW — 1

3

Explanation: The reactor must produce more power than the electrical output.

(d) Calculate the number of uranium atoms which undergo fission in 1 s. [2]

Answer

$N = \frac{4000 \times 10^{6}}{2.80 \times 10^{-11}}$ — 1

$= 1.43 \times 10^{20}$ fissions per second — 1

2

Explanation: Power ÷ energy per fission.

(e) Determine the mass of $^{235}_{92}U$ that will be consumed in 30 days. [3]

Answer

Number in 30 days $= (1.43 \times 10^{20})(30 \times 24 \times 3600) = 3.70 \times 10^{26}$ — 1

Mass $= (3.70 \times 10^{26})(235.04392)(1.66 \times 10^{-27}$ kg) — 1

Mass = 144 kg — 1

3

Total for question **15**

Explanation: Each fission uses one U-235 nucleus.